RaceDay - API Endpoint Plan

Module: PROG6212 - Programming 2B | Part 1 - System Planning and Database

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new user account as either an Organiser or Participant.	None (public)	{ fullName, email, password, role }	201 Created - user record (no password) returned. 400 Bad Request - missing/invalid fields. 409 Conflict - email already registered.
POST	/api/auth/login	Authenticates a user and returns an access token.	None (public)	{ email, password }	200 OK - { token, role }. 401 Unauthorized - invalid credentials.
GET	/api/users/me	Returns the profile of the currently logged-in user.	Any (logged in)	None	200 OK - user profile. 401 Unauthorized - no/invalid token.
PUT	/api/users/me	Updates the logged-in user's own profile details.	Any (logged in)	{ fullName, email }	200 OK - updated profile. 400 Bad Request - invalid fields. 401 Unauthorized.
GET	/api/events	Lists all upcoming events. Supports optional search/filter query params.	None (public)	None	200 OK - array of events.
GET	/api/events/{id}	Returns full details for a single event, including its categories.	None (public)	None	200 OK - event details. 404 Not Found - event does not exist.
POST	/api/events	Creates a new event under the logged-in Organiser.	Organiser	{ name, description, eventDate, location }	201 Created - new event. 400 Bad Request - invalid fields. 401 Unauthorized. 403 Forbidden - not an Organiser.
PUT	/api/events/{id}	Updates an event owned by the logged-in Organiser.	Organiser	{ name, description, eventDate, location }	200 OK - updated event. 403 Forbidden - not the owning Organiser. 404 Not Found.
DELETE	/api/events/{id}	Deletes an event owned by the logged-in Organiser.	Organiser	None	200 OK - deletion confirmed. 403 Forbidden. 404 Not Found.
GET	/api/events/{id}/categories	Lists all categories for a specific event.	None (public)	None	200 OK - array of categories. 404 Not Found - event does not exist.
POST	/api/events/{id}/categories	Adds a new category (e.g. 10km) to an event.	Organiser	{ name, distanceKm, price }	201 Created - new category. 400 Bad Request. 403 Forbidden - not the owning Organiser. 404 Not Found - event does not exist.
PUT	/api/categories/{id}	Updates a category's details.	Organiser	{ name, distanceKm, price }	200 OK - updated category. 403 Forbidden. 404 Not Found.
DELETE	/api/categories/{id}	Removes a category from an event.	Organiser	None	200 OK - deletion confirmed. 403 Forbidden. 404 Not Found.
POST	/api/enrolments	Enrols the logged-in Participant into a category.	Participant	{ categoryId }	201 Created - enrolment record. 400 Bad Request - already enrolled. 401 Unauthorized. 404 Not Found - category does not exist.
GET	/api/enrolments/me	Lists all of the logged-in Participant's own enrolments.	Participant	None	200 OK - array of enrolments. 401 Unauthorized.
GET	/api/events/{id}/enrolments	Lists all enrolments for an event (attendee management).	Organiser	None	200 OK - array of enrolments. 403 Forbidden - not the owning Organiser. 404 Not Found.
DELETE	/api/enrolments/{id}	Cancels the logged-in Participant's own enrolment.	Participant	None	200 OK - cancellation confirmed. 403 Forbidden - not the owner. 404 Not Found.
POST	/api/results	Captures a finish-time result for a participant's enrolment.	Organiser	{ enrolmentId, finishTime, position }	201 Created - result record. 400 Bad Request - invalid data. 403 Forbidden - not the owning Organiser. 404 Not Found - enrolment does not exist.
GET	/api/results/me	Returns the logged-in Participant's personal result history.	Participant	None	200 OK - array of results. 401 Unauthorized.
GET	/api/events/{id}/results	Returns all results for an event (leaderboard view).	None (public)	None	200 OK - array of results. 404 Not Found - event does not exist.

Endpoint plan covers Authentication, User Profile, Events, Categories, Event Enrolments, and Results as required. Implementation in Part 2 must match this table; any deviation must be explained in the README.